
Causal Stopping for Revision-Based Language-Model Reasoning: Theory, Audit, and Live Cost Trade-offs

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Stopping a reasoning model requires information available before its unseen con-
2 tinuation, whereas full-trajectory classification can exploit later answers. We study
3 complete response-and-revision increments and separate immediate correctness
4 drift from optimal continuation value. A finite-horizon formulation specializes
5 the Snell envelope to correctness minus charged computation; a drift crossing
6 is optimal under persistent conditional signs, but two exact counterexamples in-
7 validate weaker population or myopic arguments. We audit a 75,965-trajectory
8 revision matrix and a separate 28,888-trajectory five-step detector corpus. A retro-
9 spective stacked AUC of 0.955156 uses future information and non-nested fitting;
10 task-grouped causal-prefix GRU AUC is 0.874326 in a different evaluation. We
11 implement a causal runtime and train portable current and next-correctness pre-
12 dictors using disjoint task groups. On 100 actual Qwen2.5-0.5B generations, the
13 frozen learned policy saves 56.51% of completion tokens, with seven correct an-
14 swers versus six for the full horizon. Every task stops at two responses, and the
15 accuracy interval includes harm. Thus actual avoided generation is established,
16 while added value over fixed-two stopping, live calibration, and accuracy noninfer-
17 iority are not.

18 1 Introduction

19 Additional reasoning can repair an answer, repeat an unproductive calculation, or replace a correct
20 candidate with an incorrect revision. Final-answer accuracy conceals these transitions. A stopping
21 system must decide after an observed prefix whether a further increment has enough expected ben-
22 efit to justify its cost. This is a conditional allocation problem, not a claim that longer reasoning
23 is invariably helpful or harmful. It also differs from identifying the best step after the complete
24 trajectory has been graded.

25 We investigate repeated *complete responses*: a model emits a thought, candidate answer, and teleme-
26 try, and the next prompt includes its earlier responses. An increment is consequently a response-and-
27 revision event rather than an arbitrary token or an observation of hidden cognition. This contract
28 makes both the decision information and the avoided model calls inspectable. Its conclusions need
29 not transfer to uninterrupted internal reasoning.

30 Three distinctions motivate the study. First, corruption, insufficient cost-adjusted benefit, and re-
31 gret against a comparator are different endpoints. Accuracy can increase while net utility drift is
32 negative. Second, immediate drift omits the option of future repair, so its sign need not determine
33 an optimal stop. Third, ranking, probability calibration, and stopping utility are different statistical

34 targets. A high full-trajectory AUC cannot be inherited by a controller that cannot see those future
35 observations.

36 Our contribution is an integrated theoretical and empirical audit rather than a claim to invent the
37 Snell envelope. We state the admissible stopping problem, prove the specialized reward reduction
38 and conditional transition identity, give a sufficient persistent-sign condition, and exhibit exact fail-
39 ures of weaker arguments. We then bind two overlapping experimental corpora to explicit denomina-
40 tors, retain matched negative results, and distinguish an inadmissible retrospective diagnostic from
41 causal-prefix evaluation. Finally, we execute an observation-only controller with a serialized trained
42 predictor, separate fitting from calibration and evaluation, and report actual generation costs along-
43 side low absolute task accuracy. The learned rule’s near collapse to a fixed-two baseline is itself an
44 important negative finding.

45 **Related work.** Chain-of-thought prompting demonstrates the utility of explicit intermediate rea-
46 soning [1], while self-consistency aggregates sampled reasoning paths [7]. Outcome verification and
47 process supervision address different forms of answer assessment [2, 6]; neither alone defines an ad-
48 missible stopping time with all assessment costs charged. Work on overthinking in long-reasoning
49 models motivates economical inference [8]. Our observational unit and controlled revision prompt
50 are narrower, and we do not treat our counts as measurements of those models’ latent thoughts. The
51 benchmarks are GSM8K, MATH, ARC-Challenge and GPQA [2–5]. GRU and rotary-position ar-
52 chitectures supply reference sequence components [9, 10]. Classical finite-horizon stopping theory
53 supplies the general optimum [11, 12]; concentration and confidence-sequence theory clarify the
54 additional assumptions needed for uncertainty certificates [13, 14]. We use these results to delimit
55 empirical claims, rather than equating a fitted score with their conditional objects.

56 2 Decision model and optimality limits

57 Fix a finite horizon N and earliest permitted stop m . On a probability space, let $\mathcal{F}_t$ be the growing
58 declared decision information after response t . A causal selector emits A_t ; $C_t = \mathbf{1}\{A_t = Y^*\}$
59 denotes its correctness against the reference, with the benchmark grader replacing exact equality
60 where appropriate. Cumulative computation cost K_t is nonnegative, nondecreasing, adapted and
61 integrable. All executed probes and peers enter both the information and cost ledgers. Offline gold
62 and future responses are not runtime inputs. An admissible stopping time satisfies $m \leq \tau \leq N$ and
63 $\{\tau \leq t\} \in \mathcal{F}_t$.

64 This is a statistical information model, not a formalization of computational hardness. If the refer-
65 ence is a known measurable deterministic function of a fully revealed task, then C_t is measurable
66 and its conditional probability equals its label, regardless of a gold API field. Nondegenerate un-
67 certainty requires a latent-reference model or an explicitly coarser decision-statistic filtration. A
68 fitted posterior over prefix summaries does not condition on all mathematical semantics. No such
69 sufficiency is inferred from its AUC.

70 Put $q_t = \mathbb{E}[C_t \mid \mathcal{F}_t]$ and $G_t = q_t - K_t$. For every admissible τ ,

$$\mathbb{E}[C_\tau - K_\tau] = \mathbb{E}[G_\tau]. \quad (1)$$

71 Indeed, $\{\tau \leq t\}$ is measurable in $\mathcal{F}_t$, so multiplying C_t and q_t by its indicator preserves expectation;
72 sum over the bounded times. The candidate selector is material. Our learned and full-horizon arms
73 select the latest nonempty available answer, so a blank current response can carry an earlier answer
74 forward. The heuristic adds causal retention branches specified below.

75 **Conditional transitions.** Define repair and corruption masses $r_t = \mathbb{P}(C_t = 0, C_{t+1} = 1 \mid \mathcal{F}_t)$
76 and $s_t = \mathbb{P}(C_t = 1, C_{t+1} = 0 \mid \mathcal{F}_t)$. Set $\alpha_t = r_t/(1 - q_t)$ when $q_t < 1$ and $\beta_t = s_t/q_t$ when
77 $q_t > 0$, assigning zero on null states. With $c_t = \mathbb{E}[K_{t+1} - K_t \mid \mathcal{F}_t]$,

$$\mu_t := \mathbb{E}[G_{t+1} - G_t \mid \mathcal{F}_t] = (1 - q_t)\alpha_t - q_t\beta_t - c_t. \quad (2)$$

78 The binary increment is a repair indicator minus a corruption indicator, and the tower property
79 gives the identity. It requires neither Markov structure nor independent transitions. The hazards
80 have different eligible denominators; their unweighted difference is not accuracy drift. For correct-
81 answer value v and incorrect-answer penalty p , the reward is $(v + p)q_t - p - K_t$, and the accuracy
82 part of the drift is multiplied by $v + p$.

83 **Theorem 1** (Finite-horizon optimum). *Under the stated integrability, adaptation and bounded stop-*
84 *ping assumptions, define $S_N = G_N$ and*

$$S_t = \max\{G_t, \mathbb{E}[S_{t+1} \mid \mathcal{F}_t]\}, \quad m \leq t < N. \quad (3)$$

85 *Then S is the smallest integrable supermartingale dominating G , and $\tau_t^* = \inf\{s \geq t : S_s = G_s\}$*
86 *attains $S_t = \text{ess sup}_{\tau \geq t} \mathbb{E}[G_\tau \mid \mathcal{F}_t]$.*

87 Backward induction establishes domination and minimality. Bounded optional sampling upper-
88 bounds every admissible reward by S_t . Before first contact, the recursion holds with continuation
89 equality; the stopped process is a martingale from t and attains the bound at contact. Appendix A
90 gives complete proofs. This is the standard finite-horizon construction [11, 12], specialized to the
91 declared selector and floor.

92 Optimal continuation advantage includes an additional option:

$$\Delta_t^* = \mathbb{E}[S_{t+1} \mid \mathcal{F}_t] - G_t = \mu_t + \mathbb{E}[S_{t+1} - G_{t+1} \mid \mathcal{F}_t] \geq \mu_t. \quad (4)$$

93 Thus positive immediate drift favors continuation, but nonpositive drift alone does not establish an
94 optimal stop.

95 **Proposition 2** (Persistent-sign sufficient condition). *Let T_c be the first admissible $t < N$ with*
96 *$\mu_t \leq 0$, or N . If, almost surely, every later conditional drift stays nonpositive, then T_c maximizes*
97 *expected reward over admissible stopping times.*

98 For any bounded τ , the drift decomposition gives $\mathbb{E}G_\tau = \mathbb{E}G_m + \mathbb{E} \sum_{s=m}^{N-1} \mathbf{1}\{\tau > s\} \mu_s$. Before T_c ,
99 stopping earlier omits positive terms; afterward, continuing adds nonpositive terms. This proves the
100 comparison. Pathwise increasing q , decreasing α , increasing β , and nondecreasing predictable cost
101 are sufficient for nonincreasing drift, but are not established for the model panels. A single crossing
102 of a population curve is weaker.

103 **Two exact counterexamples.** With $q = (0, 0, 1)$ and cost 0.1 per increment, immediate drift at
104 zero is -0.1 ; myopic stopping earns zero, while stopping at two earns 0.8. The example shifts to
105 any floor. A second construction starts at $q_0 = 0.5$, observes a fair branch with $q_1 = 0.4$ or 0.6 ,
106 then has $q_2 = 1$ or 0 respectively. Both unconditional drifts are -0.1 , yet continuing once and then
107 continuing only on the first branch earns $0.65 > 0.5$. Appendix B realizes both with one common
108 reference answer; neither uses a hindsight decision.

109 **Estimation and uncertainty.** A feature-conditional predictor need not equal q_t or provide a suffi-
110 cient state for a compressed Bellman recursion. Calibration is a separate prediction objective [15];
111 marginal calibration need not imply history-conditional calibration (Appendix C). Weighted binary
112 log loss has optimum $r = w_1 q / [w_1 q + w_0(1 - q)]$, requiring correction or independent calibration
113 before probability interpretation. If simultaneous upper bounds cover μ_t , stopping on a nonpositive
114 bound controls false positive-drift stops at the coverage error; it does not prove Bellman optimality
115 or noninferiority. Probe independence must be stated given pre-probe information with a named
116 target. Post-probe drift can differ. Conditional supermartingale assumptions likewise cannot be
117 replaced by favorable unconditional means [13, 14]. We do not claim such live certificates.

118 3 Corpora, audit protocol, and trained controller

119 **Two overlapping records.** The revision matrix contains 52 model-domain cells, 13 model con-
120 figurations, 75,965 sanitized trajectories, 798,770 raw step rows and 1,948 unique tasks. Recorded
121 horizons are 8 for ARC, 10 for GSM8K/GPQA and 14 for MATH. The standardized detector se-
122 lection contains 52 cells, 13 configurations, 28,888 five-step trajectories, 144,440 rows and 2,948
123 tasks; it replaces InternLM with Qwen3.5-9B. They overlap and must not be summed. Historical
124 generation uses seed 7 at temperatures 0.1, 0.6 and 1.0, not independent generation seeds. GPQA
125 metadata and loader split provenance differ; the selected record has no SVAMP result. These details
126 limit generalization.

127 The freeze distinguishes exact local bytes and canonical LF content to handle checkout line endings.
128 Hashes bind selected records, not their truth, representativeness or every grader outcome. Primary
129 archive analyses retain stored labels; new training labels are a separately audited version. Task

resampling preserves repeated trajectories, while matched estimator contrasts use their declared 52-cell uncertainty unit. Historical environment metadata are incomplete. We provide verifiable saved-result reanalysis rather than claim bitwise reexecution on arbitrary hardware.

Causal runtime. The controller accepts one new completed observation, stores its own prefix, rejects skipped/repeated steps, and closes after stopping. It cannot receive gold, correctness, a future row or a full-sequence score. The live generator requests one response at a time and removes stopped tasks from later calls. Confidence is trusted as *observed telemetry* only after strict validation of the required JSON fields; fallback defaults become missing confidence. Reported confidence is not a calibrated posterior. Optional peers require a complete same-step roster with a barrier preceding the decision and fully charged costs; these experiments use no peers or verifier.

Both active and full-horizon arms use floor two, horizon five and the same response boundary. Learned and baseline selection use the latest nonempty answer. The heuristic freezes confidence 90, stability two, confidence drop 15 and two changes as its wobble setting; its drop branch retains the previous answer and its wobble branch retains the highest observed valid confidence. Neither retention branch fires in the reported live panels, so realized selection uses the common default. The learned rule instead stops on $\hat{p}_{\text{next}} - \hat{q}_t - 0.05 \leq 0$ after the floor. The cost is a declared utility preference, not a universal monetary rate. At the horizon no next-step head is queried.

Prefix probability fitting. We train on all 1,500 archived Qwen2.5-0.5B GSM8K-training and MATH trajectories [16]. A fixed salted hash of public domain/task identity assigns 902 tasks to training, 322 to calibration and 276 to evaluation before parser availability or outcomes. Public identities of the live 100 tasks and 20 traps are excluded. Each task’s rows remain in one role. Two unweighted logistic heads target correctness of the latest nonempty candidate currently selected and of the candidate selected by the same rule after one more increment. Train-only standardization and logistic coefficients use $C = 1$, L-BFGS and 2,000 maximum iterations. Independent one-dimensional logistic calibrators use only calibration tasks, $C = 1000$ and the same iteration cap. No evaluation/live/trap label selects a feature, cost or threshold.

The 21-feature contract uses current/prior step, parse success, answer presence, valid confidence and prefix summaries, generated-token lengths, answer changes and streak, answer/thought characters, fixed thought numeric and keyword densities, public domain and predetermined interactions. It excludes hidden states, labels, future responses and retrospective normalization. Runtime requires only the standard library: ordered coefficients, scalars and calibrators are serialized with source and task-split identities.

Reconstructing archived candidates through the actual live JSON/fallback parser produces 4,269 string disagreements among 7,500 rows. We regrade against *archived* references under the existing verifier; 43 correctness labels change. A row-level audit preserves both versions and the original corpus. Requiring all five candidates to agree would select an early prefix using later observations, so that provisional eligibility approach was rejected. All complete trajectories are retained. None is strict-valid live JSON, leaving confidence-related features without archive variation; the live prompt and emitted-EOS accounting are material transport changes.

Causality, transport and artifact validation. The split protects three distinct statistical operations: coefficient fitting, probability calibration, and final development measurement. A standardizer estimated on evaluation prefixes would breach that boundary even without reading their labels; a trajectory-agreement eligibility filter would breach it by reading unavailable future candidates. Our selected-row and serialization checks address these concrete failure modes. Current/next targets also refer to selected answers, not merely the most recently emitted string. Ignoring carry-forward selection creates a different supervised target precisely when a response is empty or cannot be parsed.

These protections do not establish transport to the JSON live prompt. The seven parse/confidence-related feature coefficients are zero in both heads because their archived inputs are constant. The deployed rule is therefore a learned function of the other supported prefix statistics, not evidence that self-reported confidence has been calibrated. Independently calibrated logistic heads can also have different approximation errors; subtracting their probabilities does not guarantee an unbiased drift or a uniformly covered conditional gain. We retain both head diagnostics and evaluate their actual stopping actions instead of calling the probability difference a certified hazard. No monotonicity constraint or Bellman transition model is fitted.

Table 1: Matched archival development utility contrasts per trajectory. Intervals are percentile re-samples of the 52 selected model-domain cells, not independent generation-seed uncertainty.

Modification relative to matched control	Mean contrast	95% interval
Empirical-Bayes hazards	+0.00781	[+0.00276, +0.01355]
Lagged logistic features	+0.00331	[+0.00062, +0.00633]
Gradient-boosted probe	−0.05694	[−0.08716, −0.03145]
Isotonic calibration	−0.06165	[−0.09168, −0.03547]

Runtime validation rejects wrong feature order, nonfinite coefficients, nonpositive scaler denominators and unsupported horizons/domains. Standard-library predictions are checked against the offline implementation. These are useful deployment consistency checks, but they verify execution of the frozen model, not its statistical correctness. The anonymous review supplement contains source/evidence documentation and sanitized review copies with original/review identities. It omits large raw archives and model weights, so it supports focused audit rather than complete historical rerunning.

Verification boundary and AI assistance. LLM coding agents contributed mathematical drafting, source audits, implementation, label-reconstruction design, and manuscript preparation. Verification includes independent finite-policy enumeration, numeric serialization checks, task-partition and prefix-invariance tests, artifact hashes and primary-source reference checks. Agents are not authors, and these checks are not a completed human author or independent reviewer attestation. Human verification and release/license review remain pending for this preparation draft.

4 Results

4.1 Population transitions and controlled contrasts

At the pooled GSM8K step-two prefix, 3,145 repairs occur among 14,818 incorrect candidates and 1,170 corruptions among 4,682 correct candidates. On the common 19,500-trajectory, 500-task panel, their difference divided by trajectory count and minus cost 0.05 is +0.0513. At prefix four, accuracy still increases by approximately 0.0373, yet net drift is −0.0127. Consequently a negative utility drift is not necessarily an accuracy decline.

Selected cells also contradict a universal early crossing. Qwen2.5-7B/GSM8K has drift +0.01933 at prefix four and −0.02533 at five; 32B/MATH has +0.01867 at five and −0.01333 at six. Each cell has 1,500 trajectories over 500 tasks. These are descriptive conditional-on-panel summaries, not per-question posterior bounds or proof of persistent conditional signs.

Table 1 retains both positive and negative matched results. More expressive probability fitting does not automatically improve a thresholded stopping policy. The summed +593.55 hazard contrast divided by 75,965 produces its reported per-trajectory effect; it is not a peer-agreement effect. The isotonic result also does not prove that calibration is inherently harmful. Estimator, sample, target and downstream policy jointly determine utility.

A paired Qwen2.5-7B/GSM8K precision contrast has 418/1,500 step-two correct answers under BF16 versus 204/1,500 under four-bit weights: 14.27 absolute percentage points, task-bootstrap interval [11.13,17.53]. This is not a relative percentage or universal quantization law. In the separate Mistral-Small-22B/GSM8K 256-versus-512-token comparison, both arms have 454 policy-loss indicators with zero discordant pairs. That null endpoint does not imply equal token sequences, time or all labels, or eliminate truncation elsewhere.

4.2 Retrospective ranking and fitted prefix probabilities

The stored task-grouped causal GRU has micro AUC 0.874326, task-macro 0.821392, domain-macro 0.810242 and worst-domain GPQA AUC 0.631253. Task AUC requires both classes; 2,679 of 2,948 groups are eligible. A causal RoPE transformer has lower micro AUC 0.863837 but slightly higher recorded step utility 0.333123 versus 0.326449. Their descriptive fold intervals overlap; no statistically unique architecture winner follows.

Table 2: Portable predictor on 276 task-disjoint archived evaluation tasks. Calibration uses a separate 322-task panel; scores are marginal development diagnostics, not certified live posteriors.

Target	Prefix rows	AUC	Brier	10-bin ECE
Current selected correctness	1,380	0.710094	0.097599	0.022227
Next selected correctness	1,104	0.692308	0.099697	0.020928

Table 3: Actual generated completion costs, including EOS. Both learned arms stop every task at two. Accuracy is low; intervals allow harm and no noninferiority margin was prespecified. The two panels have different scope.

Panel	Arm	Correct	Tokens	Saving	Prefix agreement
100 GSM8K tasks	Full horizon	6/100	22,244	—	reference
	Heuristic	6/100	21,591	2.94%	94/100
	Learned drift	7/100	9,674	56.51%	100/100
20 traps	Full horizon	1/20	4,736	—	reference
	Heuristic	1/20	4,578	3.34%	20/20
	Learned drift	1/20	2,268	52.11%	20/20

The retrospective stacked score is 0.955156 versus 0.943223 for its reduced-feature LightGBM control, with lift interval [0.010384,0.013480]. The interval concerns the difference, not the absolute stacked AUC. Bidirectional all-five-step scoring and centered smoothing consume future observations, while meta-training is not nested within reported scoring outer folds. The nominal *No Peers* control retains committee aggregates. Hence the stored lift isolates neither live correctness nor a pure peer effect. Bootstrap repetition does not repair information leakage. Moreover the 0.955156 and 0.874326 numbers are different evaluations; subtracting them cannot identify a causal effect of removing future features.

The serialized model’s results are separate from both earlier detectors (Table 2). Calibrated Brier scores improve modestly over raw values 0.098540 and 0.100677. Its held-out replay stops 275/276 tasks at two and one at three. Accuracy is 11.96% against full-horizon 10.51%; the paired task-bootstrap difference interval is [-1.45,+4.35] points. Archived completion savings are 51.29%, interval [49.31,53.09]. Accuracy equals fixed two, while cost is slightly higher. Thus little additional policy value is demonstrated. A distinct in-sample 7B/GSM8K replay saved 54.34% but lost 6.33 accuracy points; it remains a development diagnostic and is not live evidence.

4.3 Actual generation, costs and latency

We execute local Qwen2.5-0.5B with greedy CUDA generation, batch size 32, 128-token completion cap, floor two and horizon five. A cached GSM8K-test 100-task panel is fixed before collection; a separate 20-question handpicked mathematical bank covers rates, changing bases, dependent draws and answer contracts. Public prompts are separated from grading ledgers. The learned artifact is frozen before its active runs and paired with the actually executed prior full-horizon baseline linked by hashes. Reusing that baseline avoids redundant GPU work without converting active execution into saved-prefix replay. Shared-prefix agreement is nevertheless measured, not presumed from greedy decoding.

Table 3 reports actual avoided generation. The main learned completion-saving task-bootstrap interval is [54.83,58.01]%; its accuracy change of one point has conservative exact interval [-4.27,+6.21] points. This uses simultaneous 97.5% Clopper–Pearson intervals [17] for improvement and worsening probabilities, whose difference has at least 95% coverage under an iid task-pair reference model. The trap saving resampling interval is [45.86,57.17]%; accuracy-change bounds span [-19.68,+19.68] points. For that handpicked bank the iid-reference bound and resampling are descriptive, not randomized adversarial-population coverage. No accuracy noninferiority claim follows.

All 100 learned main tasks and 20 traps terminate at two, so the result cannot establish an advantage over fixed two. The original heuristic’s modest 2.94% main saving is far below a 30–40% target. Baseline strict JSON success is only 88/500 responses; confidence guarding largely forces its mean

stop to 4.85. The learned rule ignores full-trajectory scores and improves compute avoidance, but its low absolute accuracy and absent archive JSON support constrain its statistical interpretation.

The observed action equivalence is particularly informative. On these runs, every permitted first decision terminates the task; none uses later prefixes to choose between repair and corruption. A deterministic two-response budget would choose the same stopping indices on the realized histories. The model’s held-out AUC therefore cannot be interpreted as evidence of useful adaptive allocation on this live panel. A separate independently executed fixed-two arm is not reported, and the historical one-trace exception at response three does not overcome this limitation. The comparison with never-stop still measures avoided actual generation under a declared common response contract.

Only one main task changes correctness between learned and full-horizon arms. The exact paired interval accordingly includes accuracy loss despite a positive point estimate, and the trap panel is both smaller and purposively selected. More repeated timing measurements cannot narrow this task-level accuracy uncertainty or make the bank representative. Prespecifying a tolerance and powering a future study for that tolerance would change the evidentiary question; inspecting these results cannot retroactively register a margin. Similarly, unchanged aggregate accuracy for the original heuristic is not proof of accuracy preservation on a new task population.

Prompt and physical work differ from emitted tokens. Main baseline versus learned totals are 142,498 versus 43,874 input tokens, 57,856 versus 23,564 decode slots, and 1,127.05 versus 343.25 model seconds. These observed ledger quantities are not interchangeable with energy or universal speedup. Peers and verifier generation are zero. A final saved-MATH prepared-observation benchmark has 18,880 combined heuristic/control decisions over 100 problems and 20 repeats, maximum 0.4383 ms. The separate learned-main benchmark has 4,000 decisions and maximum 0.9369 ms, including both calibrated heads. These exclude model loading, generation, tokenization and peer waits, and are worst observed values rather than worst-case timing proofs.

5 Limitations, impact, and conclusion

The study establishes a causal interface and measured development trade-offs, not universal economical reasoning. Its theoretical optimality is conditional on a specified process, filtration, selector and reward; persistent conditional signs and feature-state sufficiency are unverified empirically. The live model is small, responses often fail the structured contract, prompts differ from training, and the frozen learned policy reduces to fixed two. Task groups protect fitting boundaries but do not establish untouched external development history, representative adversarial coverage or multi-seed generalization. The historical detector’s upstream evaluation dependence is retained as a limitation, not repaired by downstream bootstrap statistics.

Avoiding unnecessary calls can reduce inference work, but a controller can also terminate before repair or hide failures behind confident stable answers. Low accuracy and subgroup variation make a cost-only deployment claim inappropriate. No energy benefit is measured. A stronger confirmatory study needs unseen tasks, prespecified accuracy tolerance, several generation seeds, a frozen transported model, subgroup analysis and all prompt/probe costs. Human verification and an anonymous licensed artifact release remain pending. An anonymous review capsule is prepared, but complete public release and asset-license review are different steps. Before a confirmatory run, the decision information, candidate selector, cost units, model/parser versions, data roles, minimum budget and tie convention should be frozen together. Changing a prompt to improve JSON validity changes the observation law and requires new transport validation, rather than reuse of the present calibration claim. Subgroup auditing should distinguish benchmark, task difficulty and parser failures while keeping all held-out tasks in the declared denominator. Any higher-stakes application must evaluate the consequences of missed repair, not infer suitability from economical generation alone. The framework makes those requirements testable and exposes where attractive retrospective numbers cease to justify an online decision.

References

- [1] Jason Wei et al. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 35, 2022. <https://arxiv.org/abs/2201.11903>.
- [2] Karl Cobbe et al. Training verifiers to solve math word problems. arXiv:2110.14168, 2021. <https://arxiv.org/abs/2110.14168>.
- [3] Dan Hendrycks et al. Measuring mathematical problem solving with the MATH dataset. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 1, 2021. <https://arxiv.org/abs/2103.03874>.
- [4] Peter Clark et al. Think you have solved question answering? Try ARC, the AI2 reasoning challenge. arXiv:1803.05457, 2018. <https://arxiv.org/abs/1803.05457>.
- [5] David Rein et al. GPQA: A graduate-level Google-proof Q&A benchmark. arXiv:2311.12022, 2023. <https://arxiv.org/abs/2311.12022>.
- [6] Hunter Lightman et al. Let’s verify step by step. arXiv:2305.20050, 2023. <https://arxiv.org/abs/2305.20050>.
- [7] Xuezhi Wang et al. Self-consistency improves chain of thought reasoning in language models. arXiv:2203.11171, 2022. <https://arxiv.org/abs/2203.11171>.
- [8] Xingyu Chen et al. Do NOT think that much for $2+3=?$. On the overthinking of o1-like LLMs. arXiv:2412.21187, 2024. <https://arxiv.org/abs/2412.21187>.
- [9] Kyunghyun Cho et al. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proceedings of EMNLP*, pp. 1724–1734, 2014. <https://aclanthology.org/D14-1179/>.
- [10] Jianlin Su et al. RoFormer: Enhanced transformer with rotary position embedding. arXiv:2104.09864, 2021. <https://arxiv.org/abs/2104.09864>.
- [11] Thomas S. Ferguson. *Optimal Stopping and Applications*. UCLA electronic text, Chapters 3 and 5. <https://www.math.ucla.edu/~tom/Stopping/sr3.pdf>; <https://www.math.ucla.edu/~tom/Stopping/sr5.pdf>.
- [12] Goran Peskir and Albert Shiryaev. *Optimal Stopping and Free-Boundary Problems*. Birkhäuser, 2006. doi:10.1007/978-3-7643-7390-0.
- [13] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi:10.1080/01621459.1963.10500830.
- [14] Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, non-parametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021. doi:10.1214/20-AOS1991. <https://arxiv.org/abs/1810.08240>.
- [15] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of ICML*, PMLR 70:1321–1330, 2017. <https://proceedings.mlr.press/v70/guo17a.html>.
- [16] Qwen: An Yang et al. Qwen2.5 technical report. arXiv:2412.15115, 2024. <https://arxiv.org/abs/2412.15115>.
- [17] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi:10.1093/biomet/26.4.404. <https://academic.oup.com/biomet/article-abstract/26/4/404/291538>.

A Full derivations and finite-horizon proofs

Throughout this section, time is discrete, m, N are fixed finite integers, $(\mathcal{F}_t)_{t=0}^N$ is a filtration, and all random variables are defined on one probability space. Correctness $C_t \in \{0, 1\}$ need not be observable. Its conditional expectation $q_t = \mathbb{E}[C_t \mid \mathcal{F}_t]$ is adapted. Cost K_t is adapted and integrable; nondecreasing cost is needed for the stakes comparison but not for the Snell construction. Versions of conditional expectations can be chosen consistently outside one null set because only finitely many times occur. Policies and candidate selection are restricted to the declared information. All assertions involving inequalities between conditional quantities hold almost surely.

A.1 Stopped reward and transition identity

Lemma 3 (Observable reward reduction). *For every stopping time $m \leq \tau \leq N$, $\mathbb{E}[C_\tau - K_\tau] = \mathbb{E}[q_\tau - K_\tau]$.*

Proof. For each t , the event $\{\tau = t\}$ belongs to $\mathcal{F}_t$. Thus

$$\mathbb{E}[\mathbf{1}\{\tau = t\}C_t] = \mathbb{E}[\mathbf{1}\{\tau = t\}\mathbb{E}[C_t \mid \mathcal{F}_t]] = \mathbb{E}[\mathbf{1}\{\tau = t\}q_t].$$

Sum over $t = m, \dots, N$ and subtract the common integrable cost. No conditional independence or measurability of C_t itself is required. $\square$

Lemma 4 (Binary conditional drift). *With $r_t, s_t, \alpha_t, \beta_t, c_t$ defined as in Section 2, $r_t = (1 - q_t)\alpha_t$, $s_t = q_t\beta_t$, and Equation (2) holds.*

Proof. The conditional repair mass is at most $\mathbb{P}(C_t = 0 \mid \mathcal{F}_t) = 1 - q_t$, and the corruption mass is at most q_t . They therefore vanish on the respective denominator-zero events. The stated ratio definitions and zero conventions give the mass equalities everywhere outside a null set, with $0 \leq \alpha_t, \beta_t \leq 1$. Pointwise,

$$C_{t+1} - C_t = \mathbf{1}\{C_t = 0, C_{t+1} = 1\} - \mathbf{1}\{C_t = 1, C_{t+1} = 0\}.$$

Since $\mathcal{F}_t \subseteq \mathcal{F}_{t+1}$, the tower property gives $\mathbb{E}[q_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[C_{t+1} \mid \mathcal{F}_t]$. Subtract q_t and the conditional cost increment to obtain the identity. $\square$

For finite-sample descriptive transitions, if the same n trajectories provide all adjacent binary outcomes, let n_{01} and n_{10} be the repair and corruption counts. Their difference divided by n , rather than the difference of the two eligible-subgroup rates, is the empirical accuracy increment. This is an exact counting identity, not a population inference. An affine reward $vC_t - p(1 - C_t) - K_t$ reduces to $(v + p)q_t - p - K_t$; its drift is $(v + p)[(1 - q_t)\alpha_t - q_t\beta_t] - c_t$.

A.2 Proof of Theorem 1

The recursion in Equation (3) constructs an adapted integrable process: finite maxima of integrable variables remain integrable. It dominates G and satisfies $\mathbb{E}[S_{t+1} \mid \mathcal{F}_t] \leq S_t$. If an integrable supermartingale Z dominates G , then $Z_N \geq S_N$. Backward induction yields

$$Z_t \geq G_t, \quad Z_t \geq \mathbb{E}[Z_{t+1} \mid \mathcal{F}_t] \geq \mathbb{E}[S_{t+1} \mid \mathcal{F}_t],$$

so $Z_t \geq S_t$. This proves minimality.

Fix a deterministic start $t \geq m$. First contact $\tau_t^* = \inf\{s \in \{t, \dots, N\} : S_s = G_s\}$ exists since contact holds at N , and is a stopping time since S_s, G_s are adapted. For any bounded $\tau \geq t$, telescoping the stopped supermartingale gives

$$S_\tau = S_t + \sum_{s=t}^{\tau-1} \mathbf{1}\{\tau > s\}(S_{s+1} - S_s).$$

Each indicator is $\mathcal{F}_s$ -measurable. Conditioning first on $\mathcal{F}_s$ and then on $\mathcal{F}_t$ shows $\mathbb{E}[S_\tau \mid \mathcal{F}_t] \leq S_t$, and hence $\mathbb{E}[G_\tau \mid \mathcal{F}_t] \leq S_t$.

Before contact, $S_s > G_s$, so the recursion forces $S_s = \mathbb{E}[S_{s+1} \mid \mathcal{F}_s]$. Consequently, for $s \geq t$,

$$\mathbb{E}[S_{(s+1) \wedge \tau_t^*} \mid \mathcal{F}_s] = S_{s \wedge \tau_t^*}.$$

388 This stopped process is a martingale from the chosen start t . At terminal contact $S_{\tau_t^*} = G_{\tau_t^*}$;
 389 repeated conditional expectation yields $S_t = \mathbb{E}[G_{\tau_t^*} \mid \mathcal{F}_t]$. Every admissible conditional reward is
 390 bounded by S_t , and one admissible reward attains it, proving the essential-supremum representation
 391 and optimum. This also proves the claims for an imposed floor without permitting stops before that
 392 floor. The proof is the finite discrete specialization of Ferguson [11], Peskir and Shiryaev [12].

393 Equation (4) follows by adding and subtracting $\mathbb{E}[G_{t+1} \mid \mathcal{F}_t]$. Domination $S_{t+1} \geq G_{t+1}$ proves
 394 the nonnegative option term; no bound on its size is supplied by an AUC. If a finite measurable
 395 candidate set is available at a stop, replace the immediate reward by the maximum of its condi-
 396 tional candidate rewards. This extension requires candidate-specific estimates and does not justify
 397 selecting a candidate using unavailable gold.

398 A.3 Compensator and persistent signs

399 **Lemma 5** (Finite drift decomposition). *Define*

$$D_t = \sum_{s=m}^{t-1} \mu_s, \quad M_t = G_t - G_m - D_t, \quad t \geq m.$$

400 *Then M is an integrable martingale with $M_m = 0$, and for any admissible τ ,*

$$\mathbb{E}G_\tau = \mathbb{E}G_m + \mathbb{E} \sum_{s=m}^{N-1} \mathbf{1}\{\tau > s\} \mu_s. \quad (5)$$

401 *Proof.* Adaptation and integrability follow from finite sums. The next conditional increment of M
 402 is $\mathbb{E}[G_{t+1} - G_t - \mu_t \mid \mathcal{F}_t] = 0$. Alternatively, expand $G_\tau - G_m$ as the stopped telescoping sum;
 403 measurability of $\mathbf{1}\{\tau > s\}$ gives Equation (5) directly. $\square$

404 To prove Proposition 2, first observe that T_c is a stopping time because each μ_t is $\mathcal{F}_t$ -measurable.
 405 For every $s < T_c$, $\mu_s > 0$ and $\mathbf{1}\{T_c > s\} = 1$. For every $s \geq T_c$, the hypothesis gives $\mu_s \leq 0$,
 406 while $\mathbf{1}\{T_c > s\} = 0$. For every competitor τ , therefore,

$$(\mathbf{1}\{T_c > s\} - \mathbf{1}\{\tau > s\})\mu_s \geq 0$$

407 on each path. Sum and apply Equation (5) to both times. Equality can hold for other policies when
 408 some drifts vanish; the proposition claims optimality, not uniqueness.

409 The structural sufficient condition stated in the main text is proved by subtracting adjacent expres-
 410 sions:

$$\begin{aligned} \mu_{t+1} - \mu_t &= (1 - q_{t+1})(\alpha_{t+1} - \alpha_t)\alpha_t(q_t - q_{t+1}) \\ &\quad - q_{t+1}(\beta_{t+1} - \beta_t) - \beta_t(q_{t+1} - q_t) - (c_{t+1} - c_t). \end{aligned} \quad (6)$$

411 Each term is nonpositive under the pathwise conditions $q_{t+1} \geq q_t$, $\alpha_{t+1} \leq \alpha_t$, $\beta_{t+1} \geq \beta_t$, and
 412 $c_{t+1} \geq c_t$. Constant cost is a special case. No population-average replacement of these conditions
 413 is used.

414 **Corollary 6** (Stakes ordering for an unchanged potential process). *Let $0 < w_1 \leq w_2$, where*
 415 *$w = v + p$. Keep the potential trace law, filtration, candidate selector, floor, horizon and path-*
 416 *wise nondecreasing cost process fixed. With first-contact tie-breaking, the optimal times satisfy*
 417 *$\tau^*(w_1) \leq \tau^*(w_2)$ almost surely.*

418 *Proof.* The constant incorrect-answer offset cancels from continuation comparisons. The normal-
 419 ized advantage of being forced to continue at least once is

$$\frac{\Delta_t^*(w)}{w} = \operatorname{ess\,sup}_{\tau: t+1 \leq \tau \leq N} \mathbb{E} \left[q_\tau - q_t - \frac{K_\tau - K_t}{w} \mid \mathcal{F}_t \right].$$

420 The Snell representation proves this identity by optimizing the remaining bounded stop after the
 421 mandatory increment. Since $K_\tau - K_t \geq 0$, each expression inside the essential supremum is
 422 nondecreasing in w . So is the supremum. Strict continuation for w_1 implies strict continuation for
 423 w_2 at that same history. If the higher-stakes first contact preceded lower-stakes contact on any path,
 424 this implication at the earlier time would be contradicted. The common forced terminal stop handles
 425 N . $\square$

426 Stakes must not change the prompt, generation law, selection, or cost. The result neither asserts
427 strictly increasing accuracy nor applies automatically to estimated myopic boundaries.

428 B Common-reference counterexamples

429 **Delayed repair.** Let $Y^* \in \{0, 1\}$ have a fair prior. At times zero and one the generator outputs
430 the surely incorrect symbol 2, and at time two it outputs Y^* . Before the last response the visible ob-
431 servations reveal no reference value. The decision filtration is generated by the observed responses;
432 gold is not an input. Therefore $q = (0, 0, 1)$. With $K_t = t/10$, rewards are $(0, -1/10, 4/5)$. The
433 first immediate drift is negative, but time two is optimal. A system that instead always outputs
434 symbol 2 has the same time-zero $(q, \alpha, \beta, c) = (0, 0, 0, 1/10)$, but stopping immediately is optimal.
435 Thus an exact current posterior/hazard triple need not identify a multistep optimum. Prefixing either
436 construction by uninformative responses realizes the example at any desired minimum-step floor.

437 **Unconditional drift misses adaptation.** Give the joint signal/reference distribution the four atoms
 $\mathbb{P}(A, Y^* = 1) = 1/5$, $\mathbb{P}(A, Y^* = 0) = 3/10$, $\mathbb{P}(B, Y^* = 1) = 3/10$, $\mathbb{P}(B, Y^* = 0) = 1/5$.

438 At times zero and one the candidate is 1; the signal is revealed at time one. Its two branches each
439 have probability $1/2$, so $q_0 = 1/2$, $q_1 = 2/5$ on A , and $q_1 = 3/5$ on B . At time two the generator
440 returns Y^* on A , and the surely incorrect symbol 2 on B . Hence $q_2 = \mathbf{1}\{A\}$. With cost $t/10$,
441 the expected deterministic-time rewards are $1/2, 2/5, 3/10$; both unconditional reward increments
442 equal $-1/10$. The causal policy that continues at zero, then continues on A and stops on B , earns

$$\frac{1}{2}(1 - 2/10) + \frac{1}{2}(3/5 - 1/10) = 13/20 > 1/2.$$

443 Its decision uses only the observed signal, not the future answer or its correctness. The time-one
444 conditional drift is $1/2$ on A and $-7/10$ on B ; its average is negative. The signal creates the
445 continuation option missed by the population crossing.

446 C Prediction, calibration and uncertainty conditions

447 C.1 Compressed information and weighted loss

448 If Z_t is a causal feature of the prefix, conditional transition algebra also holds given $\sigma(Z_t)$. It need
449 not give the same conditional quantity as the full history, and the fields $\sigma(Z_t)$ at successive times
450 need not be nested. Bellman compression additionally requires that, given the compressed state,
451 the conditional future observation, cost and reward law not depend on omitted history. A correctly
452 specified partially observed Markov model can instead use its full latent-state posterior. The portable
453 logistic heads provide neither such model nor sufficiency proof.

454 For $0 < p < 1$, positive weights w_1, w_0 , and $0 < r < 1$, weighted conditional Bernoulli loss is

$$L(r) = -w_1 p \log r - w_0 (1 - p) \log(1 - r).$$

455 Its derivative vanishes uniquely at

$$r = \frac{w_1 p}{w_1 p + w_0 (1 - p)}, \quad p = \frac{w_0 r}{w_1 (1 - r) + w_0 r},$$

456 because its second derivative is strictly positive. Endpoint optima are interpreted by limits. This
457 inversion concerns an unconstrained population optimum; regularization, misspecification and esti-
458 mated weights still matter. We avoid class weights for the new heads. AUC is invariant to strictly
459 increasing score transforms, whereas probability drift thresholds are not. Even a constant $1/2$ score
460 is marginally calibrated in a balanced population whose available history fully reveals correctness.
461 It is not that history's conditional posterior.

462 C.2 Algebraic bounds and their premises

463 Let all probability arguments lie in $[0, 1]$, with genuine pointwise errors $|\hat{q} - q| \leq \epsilon_q$, $|\hat{\alpha} - \alpha| \leq \epsilon_\alpha$,
464 $|\hat{\beta} - \beta| \leq \epsilon_\beta$, and $|\hat{c} - c| \leq \epsilon_c$. Subtraction gives

$$\hat{\mu} - \mu = (1 - \hat{q})(\hat{\alpha} - \alpha) - \hat{q}(\hat{\beta} - \beta) - (\hat{q} - q)(\alpha + \beta) - (\hat{c} - c),$$

so the triangle inequality yields

$$|\hat{\mu} - \mu| \leq 2\epsilon_q + (1 - \hat{q})\epsilon_\alpha + \hat{q}\epsilon_\beta + \epsilon_c.$$

For valid rectangular intervals contained in $[0, 1]$, exact extrema are

$$L = (1 - q_U)a_L - q_U b_U - c_U, \quad U = (1 - q_L)a_U - q_L b_L - c_L.$$

The drift increases with α and decreases with q, β, c ; the indicated corners attain the extrema. For the direct two-head identity $\mu = p_{\text{next}} - q - c$, hypothetical pointwise head errors instead give $|\hat{\mu} - \mu| \leq \epsilon_{\text{next}} + \epsilon_q + \epsilon_c$. Reported Brier, ECE and AUC do not supply any of these pointwise premises.

C.3 A scoped sequential sign certificate

Assume observable upper bounds U_t satisfy $\mathbb{P}(\exists t \in \{m, \dots, N - 1\} : \mu_t > U_t) \leq \delta$. Let τ_U be the first $U_t \leq 0$, with terminal fallback. On the simultaneous coverage event, every nonterminal stop obeys $\mu_{\tau_U} \leq U_{\tau_U} \leq 0$. Thus

$$\mathbb{P}(\tau_U < N, \mu_{\tau_U} > 0) \leq \delta.$$

This is a one-step sign guarantee, not optimality or retained accuracy. Replacing the target by the full Bellman advantage would require a separately valid bound for that advantage.

To obtain a conditional Hoeffding construction, name the pre-probe field $\mathcal{H}_t$ and its measurable target θ_t . Suppose n_t is fixed before observing probes, and conditional on $\mathcal{H}_t$, probes $X_{t,i} \in [a, b]$ are independent, each with mean θ_t . Conditional Hoeffding gives

$$\mathbb{P}\left(\theta_t > \overline{X}_t + (b - a)\sqrt{\frac{\log(1/\delta_t)}{2n_t}} \mid \mathcal{H}_t\right) \leq \delta_t.$$

Taking $\delta_t = \delta/[r(r + 1)]$, $r = t - m + 1$, and summing yields simultaneous coverage of the θ_t targets. Cross-time independence is unnecessary. Within-time conditional independence and the named means remain necessary. The post-probe field includes the probes and their costs; a bound for θ_t covers post-probe μ_t only when equality or a justified comparison is established. Fixed-count Hoeffding does not license adaptively choosing the probe count without further uniformity [13, 14].

For an elementary sequential check, suppose $X_i \in [-1, 1]$ is adapted to $\mathcal{G}_i$, and the null satisfies $\mathbb{E}[X_i \mid \mathcal{G}_{i-1}] \geq 0$ at every i . For fixed $0 \leq \eta < 1$,

$$E_n^{(\eta)} = \prod_{i=1}^n (1 - \eta X_i)$$

is nonnegative and has next conditional expectation at most its current value. A fixed convex mixture also starts at one and is a supermartingale. If T is its first crossing of $1/\delta$, bounded optional sampling gives $1 \geq \mathbb{E}[E_{T \wedge n}] \geq \mathbb{P}(T \leq n)/\delta$; let n increase. Hence $\mathbb{P}(\sup_n E_n \geq 1/\delta) \leq \delta$. This tests the specified conditional-mean null. If instead $X_i = B$ for all i , with one fair shared sign, each marginal mean is zero but the ten-factor product with $\eta = 1/2$ exceeds 20 with probability $1/2$. Favorable marginal means alone therefore cannot establish a 5% guarantee. Neither sequential construction is claimed for the live fitted controller.

D Task roles, feature contract and execution details

D.1 Training and label reconstruction

The model source comprises two five-step global archives: 1,000 GSM8K-training tasks and 500 MATH tasks. The task key is public domain joined to task identity. A SHA-256 mapping with fixed salt `prefix-model-v1-task-split-fixed-20261002` and thresholds 0.6, 0.8 assigns every task before parser checks, labels or model fitting.

Domain	Train	Calibration	Evaluation
GSM8K	599	225	176
MATH	303	97	100
Total	902	322	276

Public identities for the 100 live tasks and 20 traps are excluded without reading their outcomes. Within each complete trace, the candidate for a prefix is the latest nonempty answer available there; both targets use this same causal selection, with the next target evaluated after one additional response. Gold access is confined to offline reconstruction of archived training/calibration/evaluation labels and final external grading.

The strict parser requires string thought and nonempty string answer, integer confidence in $0, \dots, 100$ excluding booleans, and boolean stop. It scans decodable JSON objects and accepts one satisfying all fields. Otherwise the existing numeric/math fallback extracts an answer, but confidence is missing. Answer-change features casefold and normalize whitespace; they do not use reference-based numeral normalization. Offline verification regrades the reconstructed answer against its stored archive reference. The audit keeps old candidate/label and reconstructed candidate/label for all 7,500 rows. The 4,269 string changes and 43 label changes do not justify a claim that every new label is correct.

The serialized feature order is given by the explicit indices:

Index	Feature name
1	step
2	parse_success
3	answer_present
4	confidence_centered
5	confidence_missing
6	confidence_delta
7	prefix_confidence_mean
8	log_generated_tokens
9	log_cumulative_generated_tokens
10	answer_changed
11	answer_streak
12	prefix_answer_changes
13	log_answer_characters
14	log_thought_characters
15	thought_numeric_density
16	thought_uncertainty_density
17	thought_verification_density
18	domain_gsm8k
19	domain_math
20	step_confidence_interaction
21	streak_confidence_interaction

Lengths use $\log(1 + x)$. Centered confidence is $(h - 50)/50$ when valid and zero otherwise, with a separate missing indicator. Consecutive confidence difference is divided by 100 and is zero when either response is invalid. Prefix mean uses only valid confidences divided by 100, with default 0.5. Streak counts consecutive equal answer keys, and changes count adjacent unequal keys. Numeric density is digit characters divided by thought characters. Lowercase alphabetic words define the other densities: uncertainty words are maybe, uncertain, unsure, however, alternatively, doubt; verification words are check, verify, recheck, calculate, confirm, verification. Zero-length denominators are replaced by one. Interactions multiply centered confidence by step or streak. None reads timing, a hidden state or future row.

Both base heads use unweighted L_2 -regularized logistic regression with $C = 1$, L-BFGS and 2,000 maximum iterations. Each train-only standardizer is reused for the calibration and evaluation panels. Calibration fits an unweighted logistic transform of that head's raw logit with $C = 1000$ on calibration tasks only. The runtime reproduces the standardized affine logit and calibrated sigmoid from finite serialized numbers, verifies schema, feature order, scales, domain and floor/horizon, and hashes exact artifact bytes. No strict JSON row exists in the archives: parse and confidence-related features are constant there and their corresponding fitted coefficients are zero. Thus their presence in the interface does not demonstrate learned calibration on live confidence. An EOS token emitted in live generation is charged; archival token counts omit that possible extra token.

534 The final model identity is

535 92fe0af86ac0f204d514a6938d0a29da
536 ce2b3cdfa24800cc4e2c85a2d51879f.

537 Its task-split byte identity is

538 d23f54010b85b3071e7ffb5c58ce5a89
539 0365e84174d19154c3efb2f19517fee0.

540 The original manifest used during training is preserved alongside the model; later master-freeze
541 refreshes do not replace this provenance object. The rejected agreement-filtered provisional model
542 is retained as a superseded development record and is not the executed artifact.

543 **D.2 Live prompt and controls**

544 The system message requests a careful mathematical solution; the user prompt contains the public
545 task, current step of five, and all previously emitted thought/answer/confidence observations. It asks
546 for one short reasoning or verification sentence and an updated current answer, using exactly one
547 JSON object with thought, answer, integer confidence, and boolean stop; the answer field contains
548 only the current final answer. History is the same observed prefix for either policy when prefixes
549 agree. Chat-template formatting is performed by the model tokenizer.

550 Each generator call ends on a valid complete JSON object, EOS, a 128-token completion cap, or
551 external cancellation. The full-horizon baseline has the same response boundary; its policy continues
552 through five responses. The heuristic requires valid confidence for confidence-based checks and
553 uses confidence 90, two equal answers, confidence drop 15 and two changes. The learned controller
554 requires two observations, then stops on $\hat{p}_{\text{next}} - \hat{q} - 0.05 \leq 0$ and otherwise continues. It forces stop
555 at five before calling a nonexistent next head. The model-reported stop field is recorded telemetry,
556 not an independent correctness oracle. The default selector carries forward the last nonempty answer.
557 The heuristic drop branch instead retains the immediately previous answer, and its wobble branch
558 chooses the highest-confidence valid observed answer, breaking ties toward the earliest step. Neither
559 branch occurs in these actual main or trap runs; learned/baseline use the default selector. No verifier
560 or peer is executed in these experiments.

561 Controllers accept only consecutive Observation objects, own the prefix, and reject an append after
562 closure. The generation adapter checks caller cancellation during decoding. Optional peer measure-
563 ments must use same-step complete rosters, reject future entries and broken barriers, and charge
564 all peer work. Tests check those API invariants; an API field restriction alone does not prove the
565 statistical filtration is sufficient. Frozen executed sources and copied predictor bytes are bound by
566 the run manifests. Greedy batched generation can change when batch composition changes, so the
567 observed 94/100 heuristic and 100/100 learned shared prefixes are explicitly recorded.

568 **D.3 Hardware, software and accounting**

569 Actual generation runs on one NVIDIA GeForce GTX 1650 with 4,096 MiB device memory, using
570 float16 CUDA/SDPA, greedy decoding, batch size 32, seed 20261002 and a local Qwen2.5-0.5B
571 snapshot. The recorded environment is Python 3.12.4, PyTorch 2.11.0+cu128, Transformers 5.8.0,
572 NumPy 2.3.5, pandas 2.2.2 and scikit-learn 1.8.0 on Windows 11. Historical generation metadata
573 are incomplete and are not replaced by these current-environment details.

574 Completion tokens are emitted tokens including EOS, distinct from padded prefill and physical
575 decode slots. Main baseline/learned values are 22,244/9,674 completion tokens, 142,498/43,874
576 prompt tokens, 231,856/58,624 padded prefill slots, 57,856/23,564 decode slots and 1,127.05/343.25
577 model seconds. Trap baseline/learned model times are 245.14/74.12 seconds. These timings ex-
578 clude model loading and are not total project compute or energy. Input/padding histories and batch
579 composition preclude deriving a universal speedup from the completion-saving percentage. Failed
580 historical runs and full development compute were not comprehensively logged. CPU fitting and
581 local source/document work are additional unmetered development work.

582 The prepared-observation heuristic/control microbenchmark covers 18,880 decisions, median
583 0.0034 ms, 99th percentile 0.0130 ms, maximum 0.4383 ms. The learned-main fea-

584 ture/head/controller microbenchmark covers 4,000 decisions, median 0.0590 ms, 99th percentile
585 0.314315 ms, maximum 0.9369 ms. Neither includes generation, model loading, tokenization,
586 telemetry extraction from raw text or peer barriers. Repeats measure the same saved problem pre-
587 fixes, not 20 independently generated task samples.

588 E Uncertainty and reproducibility boundaries

589 Archived learned replay uses 2,000 paired task-cluster percentile bootstrap replicates with seed
590 20261002. All rows of a selected task move together. Live cost intervals use 10,000 paired task
591 resamples with seed 804, recomputing $1 - \sum_i T_i^{\text{active}} / \sum_i T_i^{\text{baseline}}$ in each replicate. They are em-
592 pirical percentile intervals under resampling assumptions, not exact distribution-free finite-sample
593 guarantees, and do not include across-seed generation uncertainty. The selected 52-cell contrasts
594 use cells, not rows, as their separately declared resampling units.

595 For paired correctness, define improvement indicator $I_i = \mathbf{1}\{C_i^{\text{active}} = 1, C_i^{\text{baseline}} = 0\}$ and
596 worsening indicator $W_i = \mathbf{1}\{C_i^{\text{active}} = 0, C_i^{\text{baseline}} = 1\}$. Under an iid task-pair reference model,
597 their totals each have a binomial marginal distribution, even though I_i, W_i are mutually exclusive.
598 Let $[L_I, U_I]$ and $[L_W, U_W]$ be two-sided Clopper–Pearson intervals at confidence 97.5%, computed
599 by binomial tail inversion [17]. Each marginal interval has coverage at least 97.5%. A union bound,
600 without independence of the two margins, gives simultaneous coverage at least 95%. On that event

$$\mathbb{P}(I_i = 1) - \mathbb{P}(W_i = 1) \in [L_I - U_W, U_I - L_W].$$

601 Main learned counts $n = 100, I = 1, W = 0$ give $[-0.0427483, 0.0621192]$. Trap counts
602 $n = 20, I = W = 0$ give $[-0.196760, 0.196760]$. The latter is an iid-reference calculation for
603 a handpicked fixed bank, not coverage for a randomized adversarial population. No noninferiority
604 margin was registered, and observing no worsened task does not establish absence of harm.

605 Exact mathematical checks use rational arithmetic for conditional probability identities, class-weight
606 inversion, interval corners, persistent-sign stopping, both counterexamples and dependence failures.
607 They compare backward induction against independently enumerated causal policies on 4,374 finite
608 reward trees and check 2,187 stakes-ordering systems. Twelve mathematical tests and fourteen
609 portable-model tests passed; thirty separate runtime tests passed. Finite checks supplement rather
610 than replace the general proofs. Portable-model checks include task-role disjointness, absence of
611 future-dependent eligibility, selector-aligned labels, prefix invariance, runtime/gold separation, zero-
612 variation archive features and prediction agreement with serialized scikit-learn heads. They do not
613 verify every benchmark grading label.

614 The local reproducibility records include the training protocol, task-role CSV, label-reconstruction
615 audit, held-out prefix scores and replay rows, immutable training freeze, final model JSON, exe-
616 cuted source snapshots, live public tasks, actual observations, decisions, cost ledgers and uncertainty
617 recipes. They are documented in a prepared anonymous review capsule containing 169 selected files
618 and explicit original/review-copy identities; large raw archives and model weights are omitted. This
619 is not yet a public license-reviewed release. Public task references, model licenses, historical gen-
620 eration provenance and human author verification require review before submission. Checksums
621 establish byte identity and audit linkage, not scientific correctness or representativeness. A late
622 archive eligibility filter and a future-dependent retrospective stack were explicitly excluded from
623 causal live claims rather than relabeled as corrected experiments.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: Sections 1–5 distinguish conditional theory, retrospective diagnostics, and actual development generation; the abstract explicitly reports fixed-two behavior and inconclusive accuracy.

Guidelines:

- The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 5 and Appendices C–E discuss filtration, calibration, parser/prompt shift, low accuracy, dependence, and incomplete historical provenance.

Guidelines:

- The answer [\[N/A\]](#) means that the paper has no limitation while the answer [\[No\]](#) means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Section 2 states the decision assumptions and gives proof sketches; Appendices A–C provide full proofs and common-reference counterexamples. Finite verification checks supplement rather than replace the proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: Appendix D fully specifies the new fitted controller and live protocol, and saved records permit numerical reanalysis. Complete historical generation/training environments and all development choices are unavailable, so full independent reexecution of every historical result is not established.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility.

In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The documented anonymous review capsule provides selected sources and evidence, but omits large raw archives and model weights (Appendix E). Public open-access hosting and a complete license-reviewed release are not established.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [No]

Justification: Section 3 and Appendix D provide task roles, ordered features, optimization/calibration settings, cost, and decoding controls for the new controller. Some historical detector-training and generation metadata remain incomplete and are not inferred from current software.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Section 4 and Appendix E define paired task/cell resampling units and exact binomial-reference bounds, including their independence and handpicked-bank limitations. No accuracy noninferiority or universal timing guarantee is claimed.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: Appendix D reports the live GPU, memory, software, decoding, model seconds, and token/padding ledgers. Full historical project compute, failed runs, and additional development CPU work were not comprehensively measured.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [N/A]

Justification: A completed human author review of the NeurIPS Code of Ethics is not available for this agent-assisted preparation draft. Section 3 discloses this verification boundary; license and release review remain pending.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 5 discusses potential compute accessibility benefits and harms from premature stopping, low accuracy, and subgroup variation. It calls for confirmatory accuracy/subgroup evaluation and avoids unmeasured energy claims.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The work does not introduce or release a new high-risk foundation model or scraped image dataset. It studies public reasoning benchmarks and a small probability controller; a broader licensed artifact release remains pending.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: The benchmark, model, architecture, and theory creators are cited in the references. A complete version-by-version asset license/terms audit and release packaging are still pending, so compliance is not represented as fully verified.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: Appendices D–E specify the new model, audits, decisions, tests, and scope limits. The accompanying anonymous review capsule contains the portable artifact and protocol/report/README documentation; final license review and public release remain pending.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: No new crowdsourcing or recruited human-subject experiment is conducted; evaluation uses existing public benchmarks and a handpicked mathematical task bank.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No new crowdsourcing or recruited human-subject research is conducted. The manuscript does not invent an institutional approval or human verification attestation.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: Section 3 explicitly describes LLM coding agents' contributions to theory, auditing, implementation, label reconstruction, and manuscript preparation, and the remaining human verification. The response-generation LLM is also a core experimental component whose protocol is specified.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.